

Truth-Tracking as a Homeostatic Property Cluster: What LLMs Participate In

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Abstract

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I INTRODUCTION: A BINARY PRESUPPOSITION

Do the representations inside a large language model connect to the world? The question is usually put as a yes/no. Bender and Koller (2020) say no: a system trained on form alone, like an octopus eavesdropping on telegraph traffic, learns the statistics of language without touching what language is about. Others reply that such systems meet at least some conditions on reference, or that sensory grounding isn't required for thought in the first place (Chalmers, 2023; Coelho Mollo & Millière, 2026). The camps disagree on the verdict but share the question's shape.

Both sides assume that connecting to the world is a single capacity, something a system either has or lacks. That assumption is the mistake. Truth-tracking isn't one capacity; it's a cluster of distinct mechanisms that ordinarily operate together: perceptual contact, testimonial inheritance, inferential coherence, measurement and intervention, error-correction, and practical success. A system can take part in some of these while sitting out the rest, which is just what a language model does.

The paper's claim is that truth-tracking, the family of representational successes we pick out with the predicate *true*, is stabilised by a homeostatic property cluster (HPC) in Boyd's sense (Boyd, 1991, 1999). What makes the mechanisms a cluster and not a list is that they discipline one another: perception corrects testimony, measurement corrects coherence, and practical failure corrects them all. That mutual disciplining holds the kind together, and it's what makes the cluster's instances project.

This develops Boyd's programme rather than rivalling it. Boyd (2019) already treats reference and truth as special cases of accommodation between our signalling practices and the causal structures they answer to. The cluster view of truth-tracking applies that picture one level up. Boyd's accommodationism is the background; the contribution here is the diagnostic the cluster structure supports once a system enters some accommodation practices but not the others.

That diagnostic has to be earned, because the cluster view invites an obvious complaint. Reliabilism already traces true belief to reliable mechanisms; coherentism names inferential fit; pragmatism

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names practical success; social epistemology names testimony. Listing these and calling the bundle “homeostatic” can look like relabeling, four familiar theories in one new coat. If that’s all the view does, it adds nothing.

It does more, in three ways no single-mechanism theory can match. First and most sharply, the cluster structure predicts a characteristic failure signature: pull the stabilisers apart, and the account says in advance what should go wrong. Second, it makes conflicts between mechanisms adjudicable, by treating the degree to which they co-vary as itself evidence. Third, it predicts where truth-relevant mechanisms should co-vary tightly across domains and where they should come apart.

The language model is where the first prediction pays off. A text-trained system inherits testimony and strong sensitivity to textual coherence, but it has no perceptual contact, runs no measurements, and gets little correction from acting in the world. The cluster view says such a system should produce claims that are confident, fluent, internally coherent, faithful to corpus consensus, and yet unanchored to fact. That profile has a name: hallucination.

On this account hallucination is the predicted output of a partial cluster, not a defect to be patched piecemeal. It marks the distance between the formal linguistic competence these models plainly have and the functional competence that world-contact underwrites (Mahowald et al., 2024).

The argument runs as follows. Section 2 fixes the target: the members of the cluster are representational successes (beliefs, utterances, measurements, models, inferential outputs), not truth as an abstract object. Section 3 sets out the stabilising mechanisms and their Boydian lineage. Section 4 says why the cluster projects, locating projectibility in goal achievement under counterfactual perturbation rather than in survival alone. Section 5 makes good on the adjudication and co-variation claims, the core of the reply to relabeling. Section 6 takes objections. Section 7 develops the language-model profile mechanism by mechanism. Section 8 concludes.

2 THE TARGET: WHAT *TRUE* PICKS OUT TODO.

3 TRUTH-TRACKING AS A HOMEOSTATIC PROPERTY CLUSTER

The stabilisers of truth-tracking form a causal network in Khalidi’s sense: each mechanism is a node, the links between them are causal, and the kind is the densely connected region where the nodes converge (Khalidi, 2013, 2018). Boyd’s homeostatic property cluster is the special case in which those links are actively maintained, where the mechanisms correct one another so that the properties co-occur far more reliably than chance (Boyd, 1991, 1999). Truth-tracking has such a homeostatic core, and the connections thin toward its edges.

That picture does two jobs. The homeostatic core, everyday perception and basic coordination, is where mutual correction is tight, and the coupling there is what predicts the language model’s failure signature: remove the perceptual and verificational nodes and the maintained cluster comes apart in a specific way (Section 7).

It places the loose cases. Toward the edges, in theoretical physics, ethics, and large-number mathematics, the nodes are real but sparsely linked. Reading the kind as a network doesn’t by itself fix where its boundaries fall: Onishi and Serpico (2022) show that carving a kind from a causal graph stays relative to the chosen level of abstraction, the interest-relativity Craver (2009) found in mechanism talk. So the paper fixes the purpose: truth-tracking is individuated by representing the shared

world well enough for goal achievement under perturbation (Section 4). Relative to that purpose, its connections can thin at the edges without the kind dissolving (Khalidi, 2013).

4 WHY THE CLUSTER IS PROJECTIBLE

TODO.

5 WHAT THE KIND-STRUCTURE DOES THAT COMPONENT THEORIES DON'T

This section makes good on the claim from Section 1 that the cluster view earns its keep. The test isn't whether "truth depends on many mechanisms" sounds plausible; it's whether the kind-structure answers questions that reliabilism, coherentism, pragmatism, and correspondence can't pose from inside their own resources. Read as a causal network (Section 3), the kind has structure a single-mechanism theory cannot represent: the strength of the links between its nodes. Two questions turn on that structure; a third, the language-model failure signature, is taken up in Section 7.

5.1 ADJUDICATING MECHANISM CONFLICT

When two stabilisers disagree, say perception reports one thing and well-attested testimony another, a single-mechanism theory has no internal verdict to give. Reliabilism can only ask which source is the more reliable here, which is the question at issue. The network view adds a resource the component theories lack: the strength of the links between the nodes, how tightly the mechanisms have co-varied, is itself evidence. Where the stabilisers are densely connected and have tracked one another, a lone dissenter is probably in error; where the links are weak, no single output earns much trust.

5.2 DOMAIN-CLUSTERED CO-VARIATION

The network view also predicts a cross-domain pattern no one-mechanism theory can generate: because each component theory posits a single node, it has nothing to say about how connectivity varies from one region to another. The stabilisers should be densely linked where they've been jointly disciplined (everyday perception, basic coordination) and sparsely linked where they haven't (theoretical physics, ethics, large-number mathematics). Persistent disagreement should cluster in the sparsely connected regions.

The prediction is non-circular only if connectivity is characterised without appeal to whether the mechanisms agree, and it can't be read off the graph in a purpose-free way: Onishi and Serpico (2022) show the abstraction level and boundaries stay interest-relative. The reply is to fix the purpose rather than disown it. Holding the representational aim of Section 4 fixed, connectivity is characterised by shared selection history, institutional age, and measurement infrastructure, each measurable without checking whether the mechanisms agree, and those measures are what predict where disagreement clusters.

6 OBJECTIONS

6.1 INTEREST-RELATIVITY

One objection cuts at the kind-structure itself. Onishi and Serpico (2022) argue that Khalidi's causal-network account inherits the interest-relativity Craver (2009) found in homeostatic-mechanism talk: which causal graph you draw, at what level of abstraction, and where you set its boundaries all answer to explanatory interest, so the network never fixes a unique, interest-free carving of kinds. If that's right, truth-tracking-as-a-kind is only as objective as our purposes.

The reply concedes the premise and denies the damage. The account never needed an interest-free carving. It makes two claims, both purpose-relative and both objective: a system missing the perceptual and verificational stabilisers shows the failure signature of Section 7, and connectivity tracks selection history, institutional age, and measurement infrastructure (Section 5). Each holds once the representational purpose is fixed, and neither requires that the purpose be read off nature. Onishi and Serpico (2022) reach the same general verdict, that interest-relativity is ineliminable, and Khalidi grants domain pluralism; this account makes the operative interest explicit instead of disowning it.

7 THE LLM PAYOFF: PARTICIPATION, NOT A VERDICT
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8 CONCLUSION
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